
RUPSA REGULATED NWDT SACCO LTD · CREDIT OPERATIONS

Guarantor Split Calculator

THA_1.1 — RPIP Smart Fair-Share Allocation Engine

Algorithmic loan guarantee allocation and risk mitigation, built for SACCO credit operations.

Splitting a Guarantee Fairly, Every Time

When a member's loan needs guaranteeing, several guarantors with very different deposit balances must share the exposure. Doing that split by hand creates four recurring issues in day-to-day appraisal work:



Inconsistent

The same scenario can get a different split depending on who's doing the math that day.



Slow

Recalculating by hand — or in an ad-hoc spreadsheet — eats appraisal time on every single application.



Hard to defend

No repeatable, explainable method exists when a member questions why their share is larger.



Risky at the edges

It's easy to accidentally over-expose a small-balance guarantor relative to their savings.

From Ad-Hoc Math to a Standard Method

Before — by Hand

- Manual computation, re-derived per loan file
- No documented rationale behind the split
- Inconsistent from one credit officer to the next
- Risk of over-exposing a small-balance saver

After — THA_1.1 Engine

- Instant dual-method calculation, every time
- Same transparent logic across every file
- Exposure % per guarantor flagged automatically
- Coverage shortfall caught before submission

THE SOLUTION

One Browser Tab. Two Fair-Split Methods. Zero Cost.

- 1 Enter the numbers**

Type the required guarantee amount and up to 7 guarantors' available deposits.
- 2 Get both methods instantly**

The engine computes Equal Capped and Proportional splits side by side, live.
- 3 Watch the coverage gauge**

A real-time bar shows % of the requirement covered as figures are typed.
- 4 Catch over-exposure early**

An Exposure % column flags any guarantor tied up too tightly, per method.
- 5 No data leaves the browser**

A single static HTML file — no server, no database, no member data stored.

Inputs on the Left. Fair Splits on the Right.

Loan & Guarantor Inputs

Required Guarantee (KES)

265,000

G1

75,000

G2

300,000

G3

45,000

G4

181,000

+ Add Guarantor · up to 7

Allocation Breakdown

Guarantor	Deposit	Equal Capped	Exposure
G1	75,000	73,333	97.8%
G2	300,000	73,333	24.4%
G3	45,000	45,000	100%
G4	181,000	73,333	40.5%

Coverage Gauge

100% Covered

Schematic only — recreates the live app's layout for illustration, not a literal screen capture.

Equal Capped: Share the Burden, Protect the Small Saver

Everyone contributes an equal share of the guarantee, capped at what they actually have available.

- 1 Divide the required guarantee equally among all active guarantors.
- 2 Anyone whose deposit is below that equal share contributes their full deposit — and exits the pool.
- 3 The remaining requirement is re-divided equally among the guarantors still in the pool.
- 4 Repeat until every remaining guarantor's committed amount fits within their own deposit.

Worked Example — KES 265,000

G3 (45,000 deposit)
capped at 45,000

G1 (75,000 deposit)
settles at ~73,333

G2 (300,000 deposit)
settles at ~73,333

G4 (181,000 deposit)
settles at ~73,333

Total pledged: KES 265,000 (100% covered)

Proportional: Contribute in Line With Financial Strength

Each guarantor pledges the same percentage of their own deposit — so no one gives up a disproportionate share of their savings.

Guarantee Share = (Guarantor Deposit ÷ Total Deposits) × Loan Amount

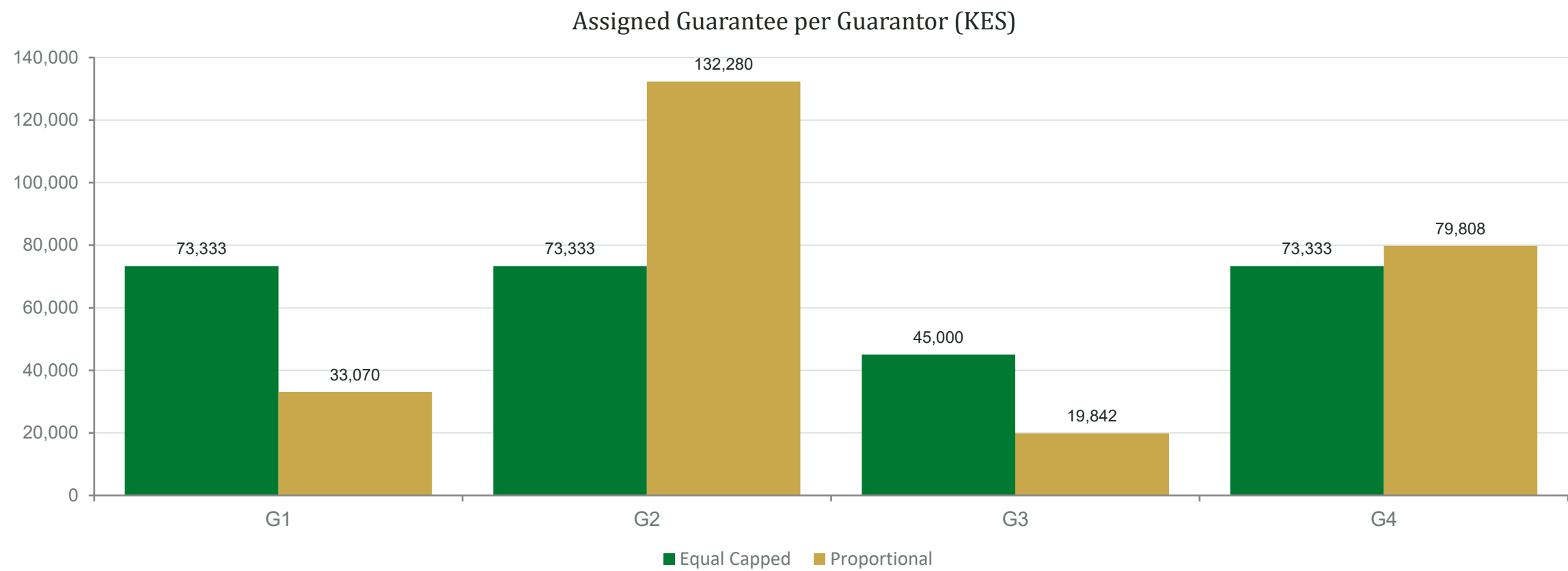
Best suited for groups where guarantors prefer contributing according to financial strength — a large-balance member absorbs more of the exposure, in KES terms, while remaining at the same percentage of their own savings as everyone else.

In the worked example (KES 265,000 requirement, KES 601,000 total pool), every guarantor pledges ~44.1% of their own deposit.

Guarantor	Deposit (KES)	% of Pool	Guarantee (KES)
G1	75,000	12.5%	33,070
G2	300,000	49.9%	132,280
G3	45,000	7.5%	19,842
G4	181,000	30.1%	79,808

Same Scenario, Two Fair Outcomes

Required guarantee: KES 265,000 · Four guarantors · Total pool: KES 601,000



Built Static. Hosted Free. Live in Minutes.



HTML5 + Vanilla JS

The entire calculation engine — no build step, no dependencies to compile.



Bootstrap 5

Layout, responsive grid, tabs, and tables — loaded from a CDN.



Animate.css

Row and card transitions for a polished, guided feel.



GitHub Pages

Free static hosting — KES 0/month, deploys straight from the repo.

Deploy in Three Steps

- Push index.html, README.md and .github/ to a new GitHub repo
- Settings → Pages → Source: main / (root) — or GitHub Actions for auto-deploy on every push
- Live at your-username.github.io/THA_1.1_Guarantor_Split_Calculator

WHAT'S NEXT

Part of a Larger Convergence

THA_1.1 is one module in the broader THA_ credit-operations series — built to eventually converge with RUPSA's wider credit-decision tooling on shared member identifiers, alongside the portfolio's risk-scoring and early-warning engines.

Thank you.